
Examples we've already seen, secretly

Some of the real life situations we've already looked at, re-focused through the lens of categories.

“Examples first” is a mantra urging us to introduce examples before theory when teaching math. However, I personally prefer seeing theory before examples, so I've always been a bit ambivalent about that mantra. Some people prefer it one way round and some people prefer it another.

Another issue with “examples first” is that it often makes some guesses or assumptions about what sort of examples will be motivating to the audience, and the reality can be wildly variable. The examples you pick might not be interesting or even familiar to your audience. If the examples are unfamiliar then it's a bit pointless doing them before the theory; rather, if the theory is interesting then it helps us understand unfamiliar examples.

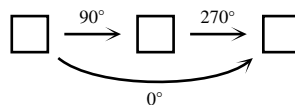
As I've said, the usual approach to category theory says that you “need” to understand a whole load of examples from mathematics before you can understand this theory that relates them all to each other. However, I am deliberately not assuming that you know any of those examples. Personally, I did know some of those examples before learning category theory, but often didn't feel fluent or familiar with them, and some of them I didn't know at all. Then when I met category theory I saw those other examples in the light of this abstract theory I really liked, and finally felt able to become familiar with them.

In this Interlude I'm going to introduce some mathematical structures as examples of categories. You might have encountered them before but I'm going to act like you haven't. A helicopter tour of a city you know is fun, as you can see familiar things fitting together at a broad scale. But it can be fun even if you don't already know the city, and it might inspire you to go back to earth and explore some things you caught a glimpse of from on high.

I'll start by going back over some of the examples we've already seen, which were “secretly” all examples of categories; later in this interlude we'll move on to more advanced mathematical examples. You could arguably skip this whole interlude and go straight into Part Two if you're not an examples sort of person; you can always come back here later if something comes up.

9.1 Symmetry

In Section 5.2 we looked at symmetry, and informally drew diagrams of arrows like this one depicting how rotations combine.



Formally we can take any shape and make a category of its symmetries. It has only one object, which we could regard as being the shape itself. The arrows are then the symmetries, and composition is given by combining symmetries in the way described in that section.

The identity arrow is the “trivial symmetry” which is technically “rotation by 0° ”. This is one of the reasons that mathematicians like considering that to be a symmetry although you might not usually do that in normal life. In normal life you’d say “this thing has no symmetry” rather than “this thing has one symmetry, the trivial symmetry”.

9.2 Equivalence relations

We briefly mentioned in the previous chapter that categories can be seen to be “inspired” by equivalence relations. I now want to stress the rigorous way of putting this: equivalence relations are a special case of categories. Suppose we are thinking about a set S with an equivalence relation $\square$ on it. Then we can make it into a category with

- objects: the elements of S
- arrows: there’s an arrow $a \rightarrow b$ whenever $a \square b$.

Things To Think About

T 9.1 Can you check that this is a category? What is the identity on each object and what is composition? How do we know it satisfies the axioms? Conversely, can you put some conditions on a category to ensure that it “is” an equivalence relation?

In the category coming from an equivalence relation the identity on each object is the relation $a \square a$, which we know is true from reflexivity. Composition comes from transitivity. The axioms are all true trivially, because in this category there just either *is* or *isn’t* an arrow $a \rightarrow b$ for any objects a and b , as an arrow is the assertion $a \square b$. The axioms for a category all declare that some arrow must equal some other arrow with the same endpoints, but there is only one arrow between any two endpoints so there is nothing to check.

That thought process gives us some clues for how to tell if some random category is in fact giving an equivalence relation. At the very least, to be able

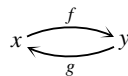
to regard arrows as assertions $a \sqsubseteq b$, there must be at most one arrow $a \rightarrow b$ for any objects a and b . This is to make sure we have a “yes/no” situation, not a “sense in which” situation. So we have no arrow, or one arrow, but not more.

The relation expressed by such a category would automatically be reflexive and transitive, from the identities and composition of the category, so it remains to ensure that it is symmetric. This means that if there is an arrow $a \rightarrow b$ then there must be an arrow $b \rightarrow a$. We are later going to see that this is the beginning of the definition of an *inverse* of an arrow.

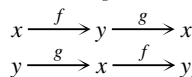
Things To Think About

T 9.2 Consider such a category, with at most one arrow from any a to b .

Consider a pair of objects with arrows going both ways.



What can we say about these two composites that result?



First note that as there can be at most one arrow from any a to b , there can in particular be at most one arrow $x \rightarrow x$ (as a and b didn’t have to be distinct). And we know there is an identity $x \xrightarrow{1_x} x$, so any other arrow $x \rightarrow x$ must be the identity. This means that the first composite above has to be the identity 1_x . Similarly the second composite has to be the identity 1_y .

We will see that this is how we define an inverse for an arrow — we ask for an arrow going “backwards” such that both of those composites are the identity. In a category coming from equivalence relations the composites are *automatically* the identity so we don’t have to specify it.

Finally here’s a table summing up how we regard equivalence relations as categories. These are deep ideas and might be difficult at first. Note the “mismatch” between what counts as data, structure and properties on each side.

equivalence relation		category	
DATA	objects	objects	DATA
STRUCTURE	relations	arrows	
PROPERTIES	reflexivity	identities	
	symmetry	inverses	STRUCTURE
	transitivity	composition	
		unit laws	PROPERTIES
		associativity	

This is a sort of dimension shift that happens quite often in category theory. As we add more nuance to situations, things that were previously properties

(often with a true/false value) become structure (“sense in which”) and then new properties are needed at a higher level to govern the structure.

9.3 Factors

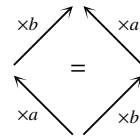
In Section 5.6 we drew some diagrams of factors which were in fact diagrams of categories. We start by picking a number, say 30 as we did in that section. Then we make a category with

- objects: all the factors of 30
- arrows: an arrow $a \rightarrow b$ whenever a is a factor of b .

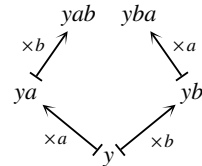
We could also do this the other way round and say there's an arrow $a \rightarrow b$ whenever a is a *multiple* of b , and the information in the diagram will be the same; it will just be depicted with the arrows pointing the other way. Turning the arrows around is called taking the *dual* in category theory, and we'll look into that more in Chapter 17.

In this category there is again a maximum of one arrow from a to b . This is because the arrows are actually coming from a relation “is a factor of” and a is either related to b or it isn't. There aren't different senses in which it might be related. It can't be related in two different ways. It's just yes or no. This means that all the diagrams commute, just like in the categories coming from equivalence relations.

Remember, a diagram is said to “commute” if all the paths through it between the same endpoints commute. In Chapter 5 we saw that the square faces in the diagrams of factors all had the form in this diagram.



If we put in the results of doing those multiplications, starting at the bottom from some number y , we can follow the two different paths around the square and see that *a priori*[†] we get the two different outcomes shown on the right.



The fact that this actually meets up in a square is then down to the fact that $yab = yba$, which is commutativity of multiplication.[‡] So in this case the “commutativity” of the diagram comes from another form of commutativity that we're more familiar with.

[†] When we say “*a priori*” we are referring to what we know in advance, from the outset, without doing any logical deduction. Usually we then follow up by deducing more things.

[‡] There's also some associativity of multiplication at work here.

We can generalize this to factors of other numbers. Given $n \in \mathbb{N}$ we can form a category with

- objects: all the factors of n
- arrows: $a \longrightarrow b$ whenever a is a factor of b .

Things To Think About

T 9.3 What would happen if we took our objects to be the whole of $\mathbb{N}$ instead?

When we do this for just the factors of n we get a shape whose number of dimensions corresponds to the number of distinct prime factors of n . If we take as objects all of $\mathbb{N}$ instead of just the factors of some fixed n we will need a dimension for every prime number, so we will have infinite dimensions.[†]

9.4 Number systems

Taking all the natural numbers as objects in the previous section resulted in something rather complicated involving infinite dimensions. That's because we were basing the structure on prime factorization. The fact that this yielded a complicated infinite-dimensional structure is somehow related to the fact that the study of prime numbers is so complicated; at least, that's how my categorical brain thinks of it.

That is not the only way to put categorical structure on numbers. It's an important part of category theory that you can take the same objects and regard them as a category in different ways depending on what you take as your arrows. The “natural numbers” are called that because they're supposed to be rather natural. But their naturalness really comes from thinking in terms of addition, not multiplication. We will look at this more in Chapter 20.

For now I want to construct much simpler categories of numbers where we base the relationships just on size rather than on factorization. Here's a way to do it for the natural numbers:

- objects: the natural numbers
- arrows: $a \longrightarrow b$ whenever $a \leq b$.

[†] This sounds like we might get an infinite-dimensional coordinate system with points corresponding to natural numbers. If you know about countable and uncountable infinities you might worry that we appear to get uncountably many points, despite the fact that $\mathbb{N}$ is countable. However, although there exist infinitely many primes, any given natural number only has a finite string as its prime factorization. So the natural numbers only correspond to points in this space where at most finitely many coordinates are non-zero.

Things To Think About

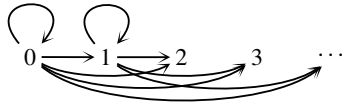
T 9.4 Try drawing a picture of this category. Remember to omit arrows that can be deduced; that is, we don't need to draw identity arrows or composites. Does it make a difference whether we include 0 or not? Could we use $<$ instead of $\leq$? What if we used $\geq$ instead?

If we think about all the arrows in this category we know that we need arrows like these. $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots$

But we also know $0 \leq n$ for all natural numbers, so we need all the arrows shown here; observe the “loop” depicting $0 \leq 0$, which is the identity arrow.



But also $1 \leq n$ for all natural numbers except 0, so we need all these arrows.



Evidently this approach is going to make a huge mess, so that turned out to be not a very helpful way of thinking about it. The thing is to remember to omit arrows we don't need as they can be deduced: identities (the “loops” above) and composites. This “tidies up” the diagram to look like this instead:

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow 5 \longrightarrow \dots$$

This diagram doesn't just look tidier, it also better matches our usual intuition about the natural numbers and how they sit on a line.

Note that it doesn't really matter whether or not we include 0 — the general shape of the category will be the same, with a starting point on the left, and then everything pointing to the right and going on forever. There's no ending point as there is not a largest natural number. We'll come back to that feature when we're talking about universal properties in Chapter 16.

We couldn't use $<$ instead of $\leq$ to make a category of natural numbers because then there wouldn't be any identities. That's because for any number a , it is *not* true that $a < a$. However, we could use $\geq$ instead of $\leq$. This just means all the arrows would point the other way:

$$0 \longleftarrow 1 \longleftarrow 2 \longleftarrow 3 \longleftarrow 4 \longleftarrow 5 \longleftarrow \dots$$

Later we will talk more about the idea of turning all the arrows around in a category. If we turn all the arrows around but leave everything else the same it's called the *dual* category. It hasn't really changed the structure of the category, it's just presented it differently. The first version and the dual version above encapsulate the same information. This small and apparently irrelevant flip in perspective actually turns out to be rather profound and often useful; we'll come back to it in Chapter 17.